

● MEDIUM

● PROBLEM-SOLVING AND DATA ANALYSIS

Percentages

10 questions · ~15 min

01

Q1

GRID-IN

PROBLEM-SOLVING AND DATA ANALYSIS

An insurance company offers a series of three information sessions. 1,250 people attended the first information session. 72% of the people who attended the first information session attended the second information session, and 36% of the people who attended the first and second information sessions attended the third information session. How many people attended all three information sessions?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q2

GRID-IN

PROBLEM-SOLVING AND DATA ANALYSIS

In a sample, 80% of the items are faulty. There are 88 faulty items in the sample. How many total items are in the sample?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

The population of City A increased by 7% from 2015 to 2016. If the 2016 population is k times the 2015 population, what is the value of k ?

- A. 0.07
- B. 0.7
- C. 1.07
- D. 1.7

Thomas installed a new stove in his restaurant. At the time of installation, the stove had a value of \$800. Thomas estimates that each year the value of the stove will depreciate by 20% of the previous year's estimated value. What is the estimated value of the stove exactly 2 years after Thomas installed it?

- A. \$480
- B. \$512
- C. \$556
- D. \$640

During the first month of sales, a company sold 1,300,000 units of a certain type of smartphone. During the same month, 15% of the units sold were returned. If sales and the return rate remain the same for each of the next 5 months, about how many units of this smartphone will be returned to the company during this 6-month period?

- A. 195,000
- B. 975,000
- C. 1,170,000
- D. 6,630,000

The value of z is 1.13 times 100. The value of z is what percent greater than 100?

- A. 11.3
- B. 13
- C. 130
- D. 213

Q7

GRID-IN

PROBLEM-SOLVING AND DATA ANALYSIS

What number is 40% greater than 115?

STUDENT-PRODUCED RESPONSE

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$\cdot$ $\frac{\cdot}{\cdot}$ $\frac{\cdot}{\cdot}$ $\cdot$

0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Which expression represents the result of increasing a positive quantity w by 43%?

- A. $1.43w$
- B. $0.57w$
- C. $43w$
- D. $0.43w$

The table summarizes the number of objects in each group.

Group	Number of objects
A	375
B	54
C	690
D	81
Total	1,200

The number of objects in group C is $p\%$ of the number of objects in group A. What is the value of p ?

STUDENT-PRODUCED RESPONSE

○	/○	/○	○
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q10

PROBLEM-SOLVING AND DATA ANALYSIS

Which of the following represents the result of increasing the quantity x by 9%, where $x > 0$?

- A. $1.09x$
- B. $0.09x$
- C. $x+9$
- D. $x+0.09$